

HEMCHANDRACHARYA NORTH GUJARAT UNIVERSITY, PATAN

B. Sc. PHYSICS :SEMESTER -V

TYPE OF COURSE: SKILL ENHANCEMENT COURSE (SEC) - THEORY

COURSE NAME: OPTICAL INSTRUMENTS

PROGRAMME CODE: SCIUG103

COURSE CODE: SC23SECPHY506

(Effective from June 2025 Under NEP-2020)

Total Credits:02	THEORY SEC	External Marks (SEE) - 25
Teaching Hours per Week:		Internal Marks (CCE) - 25
02Teaching Hours per Semester: 30		

Course Objective:

- To gain deep knowledge of principle, construction, working process of M I.
- To provide hands on experimental aspect to measurement of wavelength and determination of thickness of thin transparent sheet as well as refractive index.
- To develop knowledge of B.C. as well as CRO and usage of it.
- To learn the usage of GM counter and working function of it.

Course Outcome: After the successful completion of the course students will be able to Understanding of,

- Principle, construction, working process of M I.
- The technique of measurement of wavelength and determination of thickness of thin transparent sheet as well as refractive index.
- The principle, working function and usage of B.C. as well as CRO.
- The usage of GM counter and working function of it.

Syllabus

UnitNo.	Content	Credit	Lect. Hrs 30
Unit-1	Michelson's Interferometer (15.7) Principle, Construction, Working, Circular fringes, Localized fringes, White light fringes, Visibility of fringes(15.7.1 to 15.7.7), Applications of Michelson Interferometer (15.8)-Measurement of wavelength, Determination of difference in the wavelengths of two waves, Thickness of a thin transparent sheet, Determination of the refractive index (15.8.1 to 15.8.4) Babinet Compensator (20.21):Construction (20.21.1), Production of polarized light (20.21.2), analysis of elliptically polarized light (20.21.3). <i>Related Examples, Problems & Short Questions</i>	1	15
Unit-II	C.R.O.:CR Tube (3.5), Electrostatic Deflection Sensitivity (3.5.1), Magnetic Deflection Sensitivity (3.5.2), CRT connections (3.5.3), Uses of C.R.O.(3.5.4) G. M. Counter: Principle, Construction, Working, Dead time, recovery time, True counting rate, Efficiency of counting, Quenching of G M counter, Operation and testing of G.M. counter, Plateau, Applications of GMC, Advantages and limitations of GMC. Basic references: 1. A Textbook of Optics by Dr. N. Subrahmanyam, Brijlal&Dr.M. N.Avadhanulu, S.Chand {for M.I, B.C.} 2. Hand Book of Electronics by Gupta and Kumar. 30th revised Edition 2002. {for CRO} 3. Refresher Course in Physics Vol-III, S. Chand & Co. Ltd.(7th edition-2006) {for GMC, Ch-28}	1	15